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2015.0 RANGE ROVER (LG), 303-08

ENGINE EMISSION CONTROL - TDV8 4.4L DIESEL

PRINCIPLES OF OPERATION

For a detailed description of the engine emission system and operation, refer to the relevant Description and Operation section of the Workshop Manual.

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Engine breather hoses ▪ Oil separator ▪ Exhaust gas recirculation (EGR) pipes (check for cracks) ▪ EGR valve(s) ▪ EGR cooler(s)	▪ Fuse(s) ▪ Wiring harness ▪ Loose or corroded electrical connector (s) ▪ Intake manifold tuning (IMT) valve ▪ EGR valve(s) ▪ Engine control module (ECM)

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Difficult to start	EGR valve(s) stuck open	- Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 diagnostic test - Exhaust gas recirculation valve. Clear the DTCs and retest - Check the EGR valve and circuits. Refer to the electrical guides. Check the mechanical condition of the EGR valves. Rectify as necessary.
Poor/Erratic idle		
Lack of power when accelerating		
Engine stops /stalls	- EGR valve(s) stuck open - Breather system disconnected /restricted /blocked	- Using the manufacturer approved diagnostic system, perform routine - Inline diagnostic unit 2 diagnostic test - Exhaust gas recirculation valve. Clear the DTCs and retest - Check the EGR valve and circuits. Refer to the electrical guides. Check the mechanical condition of the EGR valves. Check the engine breather system. Check the oil separator. REFER to: Crankcase Vent Oil Separator (303-08E Engine Emission Control - TDV8 4.4L Diesel, Removal and Installation). Check for DTCs indicating an EGR valve, throttle or sensor fault. Rectify as necessary.
Excessive fuel consumption	EGR valve(s) stuck open	
Excessive black smoke	EGR not operating	
Excessive emissions	Breather system restricted /blocked	
Excessive blow-by	Breather system restricted /blocked	Check the engine breather hoses. Check the oil separator. REFER to: Crankcase Vent Oil Separator (303-08E Engine Emission Control - TDV8 4.4L Diesel, Removal and Installation). Rectify as necessary.
Engine oil leaks	Breather system restricted /blocked	Check the engine breather hoses. Check the oil separator. REFER to: Crankcase Vent Oil Separator (303-08E Engine Emission Control - TDV8 4.4L Diesel, Removal and Installation). Rectify as necessary.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: Diagnostic Trouble Code (DTC) Index - TDV8 4.4L Diesel, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation).

